Day 4 Data Report

Multi-sensor self-monitoring, 20 October 2026,
with comparison to days 1–3 and to participant 1

Participant: Spriha (participant 2)

Analysis generated with `pipeline/daily_analysis.py`,
`compare_days.py`, `compare_participants.py` and
`day_4/extra_figures.py`

1 Summary

Day 4 was spent at home. It contains the bedroom night (01:26–08:28), about five hours indoors at midday and in the afternoon (12:19–17:20), an evening indoors (17:43–21:03), four logged drinks and the wrist band, which was worn for only 6.7 of the 24 hours. **The insoles recorded nothing**: both stayed in "reconnecting" in all three recordings, so there is no gait, left-right or rickshaw analysis for this day. Table 1 lists the headline metrics. The main findings are:

- **Drinking**: 1,550 mL logged, her highest so far (599 and 1,075 mL on days 2–3), and for the first time most of it in the morning: 25 mL at 09:20 and 1,000 mL at 11:16, then nothing for 8.8 hours until 500 mL at 20:07 and 25 mL at 22:43.
- **A closed evening room** (Section 5): from 18:40, CO_2 rose steadily from about 410 to 795 ppm (146 ppm/h) while $PM_{2.5}$ fell from 65 to 50 $\mu g/m^3$, which is what happens when an occupied room is closed. Twice (18:35–18:47 and from 20:20) temperature and absolute humidity fell together without any fresh air arriving, which suggests air conditioning.
- **Environment-node warm-up**: when the node started from cold, its temperature reading rose by 10–12 °C in 30 minutes, far more than the −5 °C correction assumes. The day's corrected temperatures (34–39 °C) are therefore probably too high by a few degrees.
- **Bedroom**: $PM_{2.5}$ 53–65 $\mu g/m^3$ all night, the lowest of her two recorded nights (69 on day 3); quiet (48 dBA). The band recorded sleep from 01:41 to 05:08 (3.5 h). Heart rate was higher in the band's "deep" segment (75 bpm) than in light sleep (69), so its stage labels are doubtful.
- **Heart rate**: 86 bpm on average while awake and sitting indoors (IQR 83–90), lower than on days 1–3 (94–96), which included outings. The band was off from 05:05 to 12:19 and from 16:40 to 23:20.

Table 1: Headline metrics of day 4 (20 October 2026).

Domain	Metric	Value	Unit
Sleep	Time asleep	3.5	h
	Sleep efficiency	100	%
	Light / deep / REM	79 / 21 / –	%
	Mean HR asleep	70	bpm
	Mean SpO_2 asleep	–	%
Activity	Steps (band / phone)	131 / –	
	Mean awake HR	86	bpm
	Minutes with HR $>$ 120 bpm	0	min
	HR–steps Spearman ρ	0.17	
	HR lag behind steps	1	min
Gait (walk)	Cadence	–	steps/min
	Stride time / CV	–	s / %
	Stance, IMU (R / L)	–	%
	Swing, IMU (R / L)	–	s
	Stance, pressure (R)	–	%
	Push-off timing (R / L)	–	%
	Distance / speed	–	m / m/s
	Stride length	–	m
	Insole − band bias	–	steps/min
Posture	Monitored time	–	min
	Upright / slouched	–	%
	Median time between changes	–	min
Environment	Indoor CO_2 (median)	382	ppm
	Indoor / outdoor $PM_{2.5}$	54 / –	μg/m^3
	Indoor temp. / RH	33.5 / 41	°C / %
	Indoor / outdoor sound	60 / –	dBA
	Bedroom CO_2 / $PM_{2.5}$	356 / 57	ppm / μg/m^3
	Bedroom temp. / sound	33.8 / 48	°C / dBA
Left/right	Left lag behind right	–	% of cycle
	Foot-tilt range L − R	–	°
	Stance / swing SI (IMU)	–	%
	Stance % (L / R, IMU)	–	%
Hydration	Total intake (drinks)	1550 / 4	mL
	Longest gap between drinks	8.8	h

2 Data collected

Table 2: Day 4 recordings (local time, UTC+6).

Key	Session	Time	Sensors	Notes
O	Bedroom night	01:26–08:28	Environment	Recording ran to 12:17; node lost at 08:28
I	Indoors, midday and afternoon	12:19–17:20	Environment, GPS	Insoles never connected
E	Indoors, evening	17:43–21:03	Environment	Node disconnected 10 times after 21:03

All three uploaded folders (`Night_4`, `Day_4`, `Afternoon_4`) contain environment data only; neither insole connected at any point. The bedroom recording kept running until 12:17, but the node stopped sending at 08:28, so the night is analysed to 08:28. The evening node dropped out repeatedly after 21:03, so the evening is analysed to 21:03. The band (`Fit_2026-10-20`) recorded heart rate every minute when worn: 402 minutes in all (01:27–05:05, 12:19–14:20, 15:40–16:40 and a few minutes at 23:20). It counted only 131 steps, which reflects how little it was worn rather than her activity. GPS gave nine fixes at 14:43–14:45 that wander over about 60 m while she was indoors. This is indoor multipath, not a walk, so the route plot is left out. Drinks are from the smart-bottle log: 25 mL at 09:20, 1,000 mL at 11:16 (both in the morning), 500 mL at 20:07 and 25 mL at 22:43. Temperatures include the −5 °C correction (but see Section 5).

3 Bedroom, night of 19–20 October

$PM_{2.5}$ in the bedroom (Fig. 1) was 53–65 $\mu g/m^3$ all night (median 57). It dipped slightly after 02:00 and rose to 65–70 after 07:30. That is lower than on day 3 (69) and has no early-night peak. The room was quiet: 44–45 dBA until 03:25, when the level stepped up to 48–50 dBA (probably a fan switched on) and stayed there. CO_2 stayed at 330–380 ppm. The sensor reads low, so only changes matter, and there was no overnight build-up. The temperature curve is dominated by the node's warm-up: it rose from 26.4 to 39 °C in the first half hour and then fell slowly to 32–33 °C (Section 5). Over the same period humidity moved almost exactly opposite to temperature ($\rho = -0.97$), the signature of a sensor heating itself.

The band recorded sleep from 01:41 to 05:08 (3.5 h; Fig. 2). It reported 21 % deep and 79 % light sleep and no time awake, hence the nominal 100 %

efficiency. Heart rate averaged 70 bpm (lowest 10-minute mean 66 bpm), the same as on days 2–3 (72 and 70). In the band's "deep" segment (02:39–03:22), however, heart rate was higher (median 75 bpm) than in light sleep (69 bpm). Deep sleep normally has the lowest heart rate, so the band's stage labels should not be relied on. She appears to have got up at about 05:08: the band's record ends at 05:05, and at 05:10 the node shows a brief CO_2 spike to 560 ppm (someone breathing close to it), a sound spike to 62 dBA and a 1 °C drop in temperature. If she went back to sleep after that, the band did not record it.

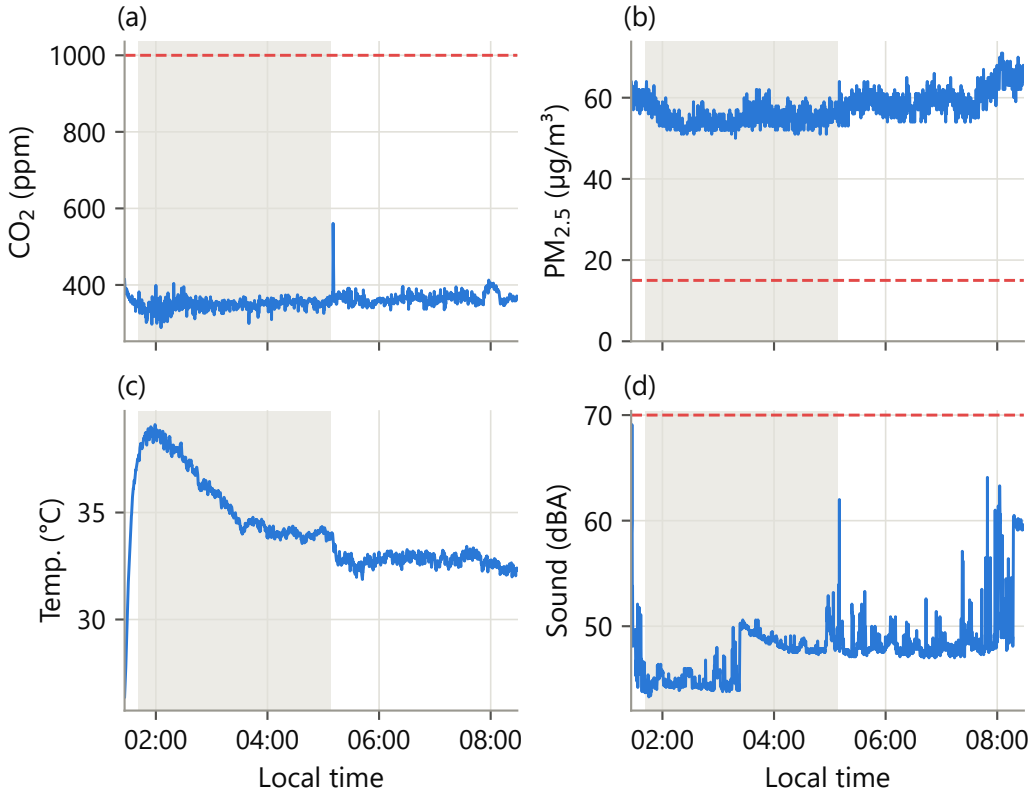


Figure 1: Bedroom environment overnight (session O, 01:26–08:28, 10 s averages): (a) CO_2, median 356 ppm (max 561); (b) $PM_{2.5}$, median 57 μg/m^3 (max 71); (c) temperature, median 33.8 °C; (d) sound level, median 48 dBA. Dashed lines: 1000 ppm CO_2, WHO $PM_{2.5}$ 24-h guideline, 70 dBA. The grey band marks sleep (01:41–05:08) from the wrist band. Temperature includes a -5 °C correction for sensor self-heating.

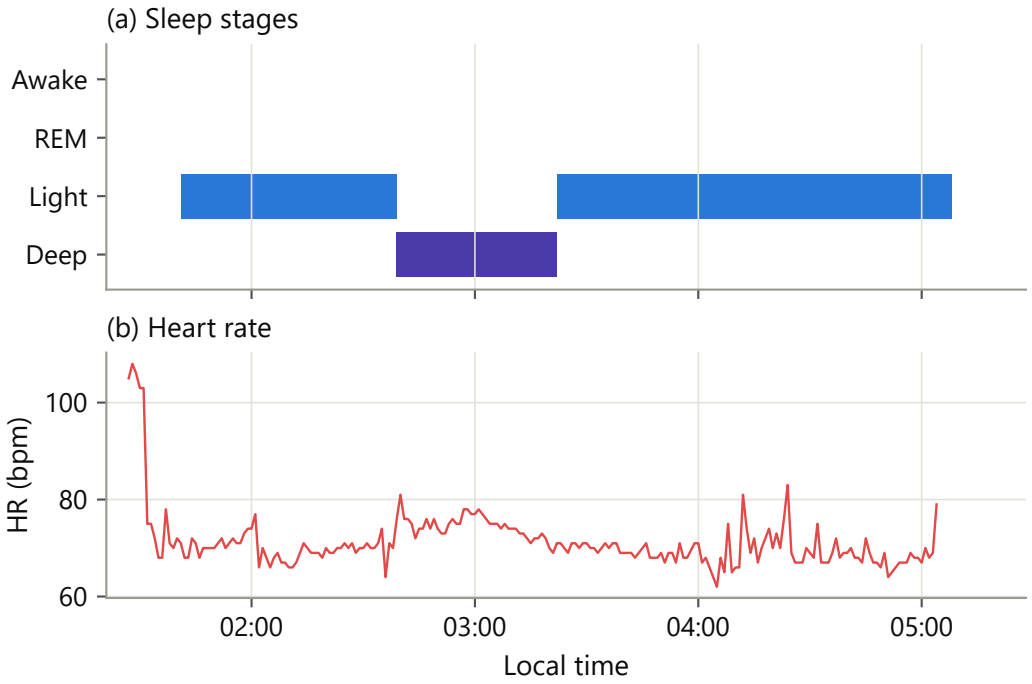


Figure 2: Sleep before day 4 from the wrist band. (a) Hypnogram, 01:41–05:08: 3.5 h asleep (efficiency 100 %), 79 % light, 21 % deep and 0 % REM. (b) Heart rate (mean 70 bpm, lowest 10-min mean 66 bpm). The band recorded no SpO_2.

4 Indoors, midday to evening

From 12:19 to 17:20 (Figs. 3–5), $PM_{2.5}$ was 50–60 $\mu g/m^3$, rising to 62–70 from 16:40. It stayed at 62–66 until 18:35, then fell once the room was closed (next section). CO_2 was 340–400 ppm, with two short rises to 440–510 ppm (13:50–14:15 and 16:50–17:05), when someone was close to the node. The sound level sat on a steady floor of 60 dBA, with sharp peaks up to 86 dBA (16:55) and four quiet spells of 48–55 dBA (14:20–14:32, 15:56–16:28, 17:51–17:57 and 18:36–18:47). A constant floor that switches off and on like this is most likely a fan or another appliance; only 0.6 % of the time was above 70 dBA. The corrected temperature was 33–37 °C after warm-up, and humidity was 34–46 %. The outdoor weather at 14:43 was 32.5 °C and 46 %.

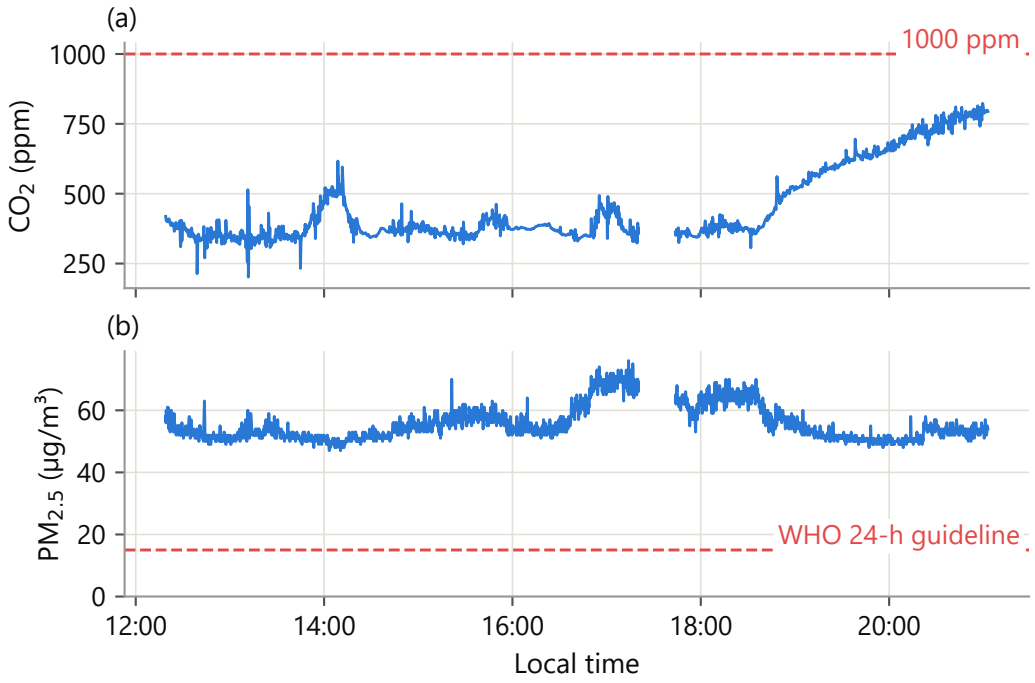


Figure 3: Indoor air quality from the environment node (10 s averages; sessions I, E, 12:19–21:02). (a) CO_2, median 382 ppm, above 1000 ppm for 0.0 % of the time. (b) $PM_{2.5}$, median 54 μg/m^3, above the WHO 24-h guideline (15 μg/m^3) for 100 % of the time. Line breaks mark periods when the node was disconnected.

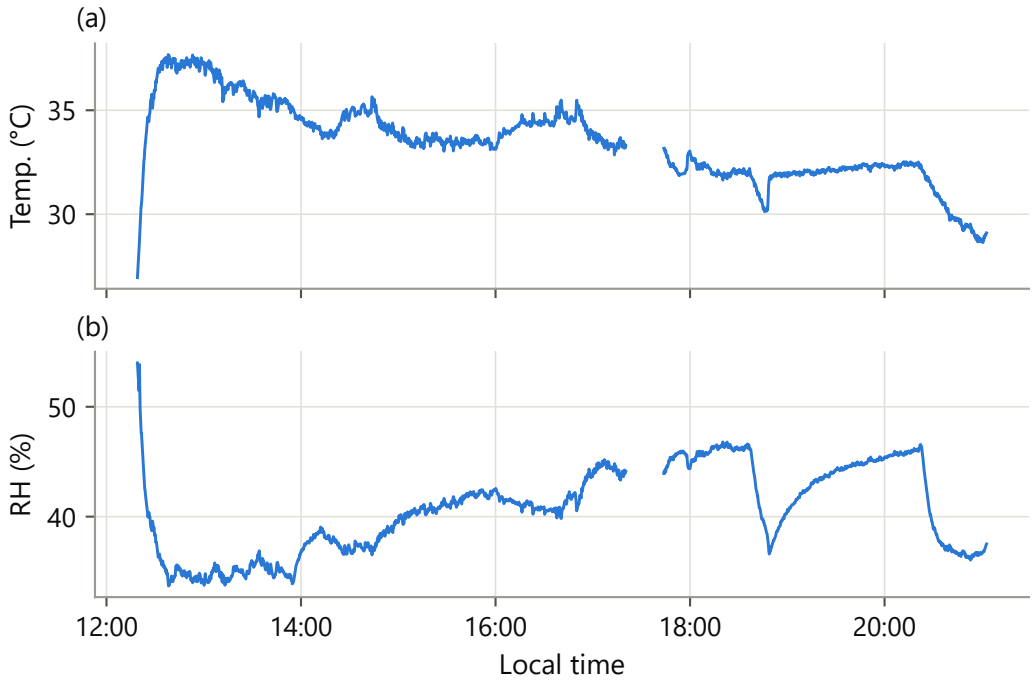


Figure 4: Indoor thermal conditions (sessions I, E, 12:19–21:02): (a) air temperature, median 33.5 °C (5–95th percentile 30.0–37.1 °C); (b) relative humidity, median 41 %. Temperature includes a -5 °C correction for sensor self-heating.

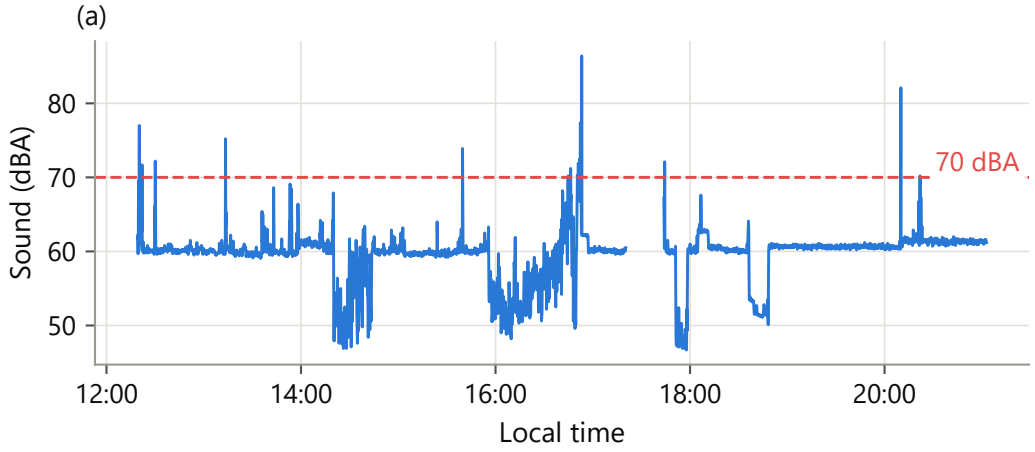


Figure 5: Indoor A-weighted sound level (sessions I, E, 12:19–21:02), median 60 dBA (95th percentile 62 dBA); above 70 dBA for 1 % of the time.

5 Interesting findings

1. The evening room was closed, then cooled and dried (Fig. 6).

- *18:35–18:47*: the fan noise stopped (51 dBA) while temperature fell by about 2 °C and absolute humidity fell from 20.4 to 15.7 g/m^3. When the fan sound returned at 18:47, temperature jumped back at once.
- *18:40–21:03*: CO_2 rose steadily from about 410 to 795 ppm (146 ppm/h) and absolute humidity climbed back from 15.7 to 20.6 g/m^3 by 20:15. Exhaled breath adds both CO_2 and water vapour, so they rise together in an occupied room with little fresh air. Over the same period $PM_{2.5}$ fell from 65 to 50 μg/m^3: with less outdoor air entering, indoor particles settle and are not replaced.
- *From 20:20*: temperature fell from 32.4 to 28.9 °C and absolute humidity from 20.6 to 13.7 g/m^3, while CO_2 kept rising at the same rate. The room was cooled and dried without fresh air, which is what an air conditioner does. The short 18:35–18:47 episode has the same signature.

CO_2 stayed below 1,000 ppm, but it was still rising when the recording ended. If the room stayed closed overnight, it would probably have passed that level. A note of when doors or windows are closed and when the air conditioner is on would confirm this interpretation.

2. The node heats itself after power-on much more than the correction assumes (Fig. 7). The bedroom and midday recordings began with uncorrected readings of 31.5 and 32.0 °C. That is close to room air, since the node had been off, and close to the 32.5 °C outdoors that afternoon. Within 30 minutes the readings had risen by 12.0 and 9.9 °C and they stayed high. The evening recording started only 23 minutes after the midday one ended, with the node still warm, and shows no rise. Across all 20 recordings of both participants, the median rise after 30 minutes was 2.5 °C, but some rose by 5–11 °C, so the self-heating varies with where the node sits and how it is ventilated. Two consequences follow:

- The first 20–30 minutes of temperature and humidity in each recording are a warm-up transient and should be excluded.
- On day 4 the warmed-up node read about 10 °C above its starting point, so the −5 °C correction leaves the indoor temperatures (corrected medians 33.5–33.8 °C) probably several degrees too high. Changes within a recording remain usable. Absolute humidity is unaffected because it is computed from the node's own temperature.

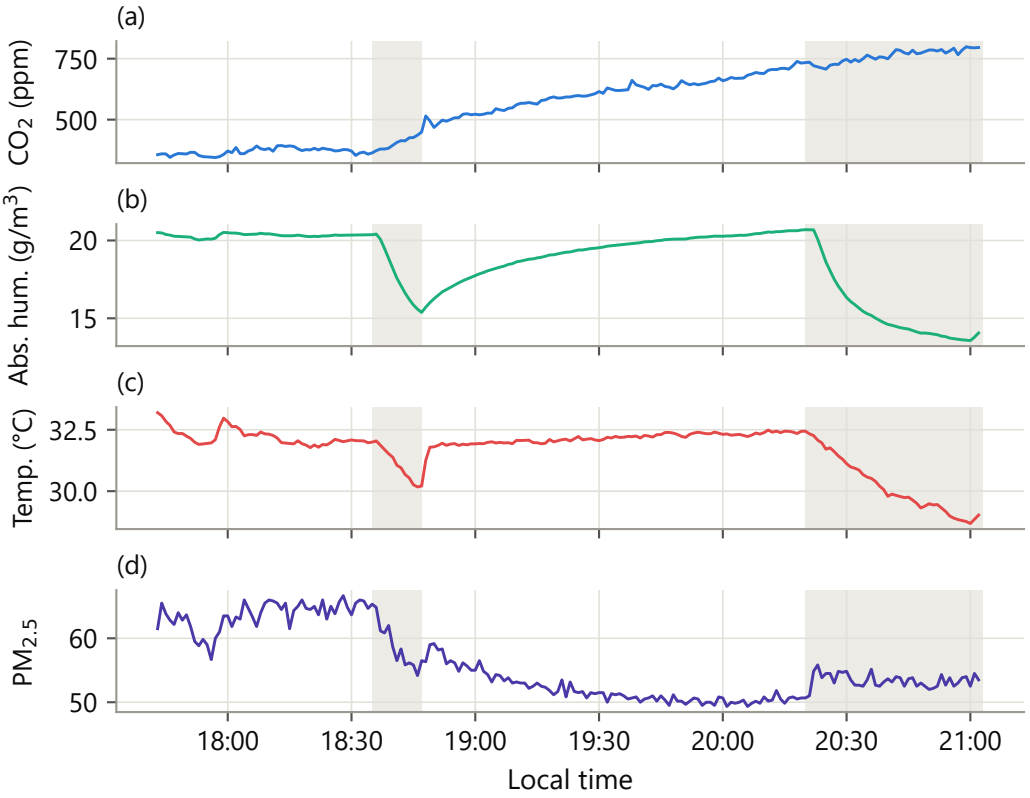


Figure 6: The evening room, 17:43–21:03 (1-min means): (a) CO_2, (b) absolute humidity, (c) temperature (corrected) and (d) $PM_{2.5}$ ($\mu g/m^3$). Grey: the two periods in which temperature and absolute humidity fell together (18:35–18:47, and from 20:20). From 18:40 CO_2 rose steadily (about 146 ppm/h) while $PM_{2.5}$ fell, the signature of a closed, occupied room.

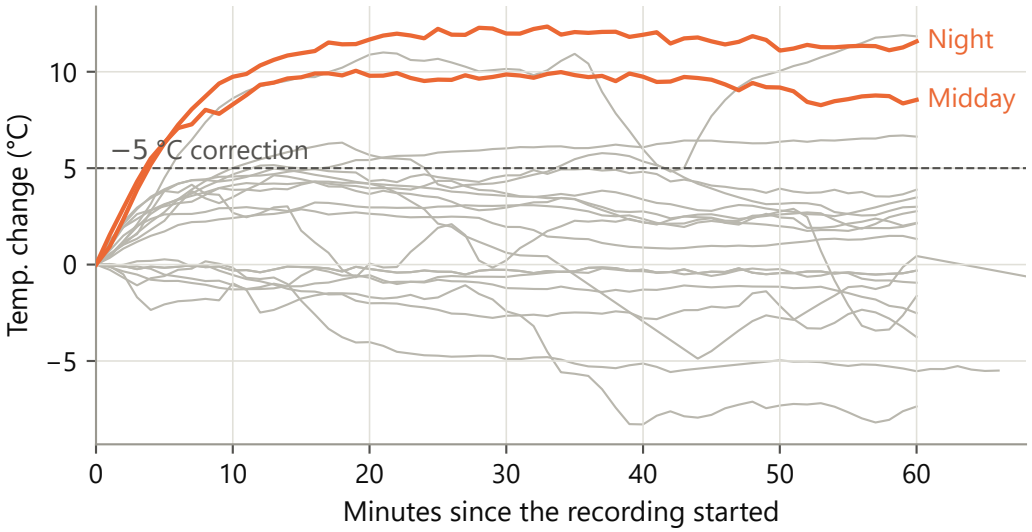


Figure 7: Environment-node temperature (uncorrected) over the first hour of every recording of both participants (20 recordings, grey), relative to its first minute. The two day-4 recordings that started from cold (orange) rose by 10–12 °C within 30 minutes; the median rise of all recordings after 30 minutes was 2.5 °C. Recordings that began with the node already warm fall or stay flat.

6 Heart rate

The band was worn for 199 awake minutes, almost all while sitting indoors at midday (Fig. 8). Awake heart rate averaged 86 bpm (68–108 bpm; no minute above 120 bpm). By context (Fig. 9) the median was 69 bpm in light sleep, 75 bpm in the band's "deep" sleep and 86 bpm sitting indoors (IQR 83–90). That is lower than on days 1–3 (94–96 bpm awake), which all included outings and walking, while day 4 was spent sitting at home. Only 3 of the 199 awake minutes contained steps, so the day's heart rate–step correlation ($\rho = 0.17$) carries almost no information.

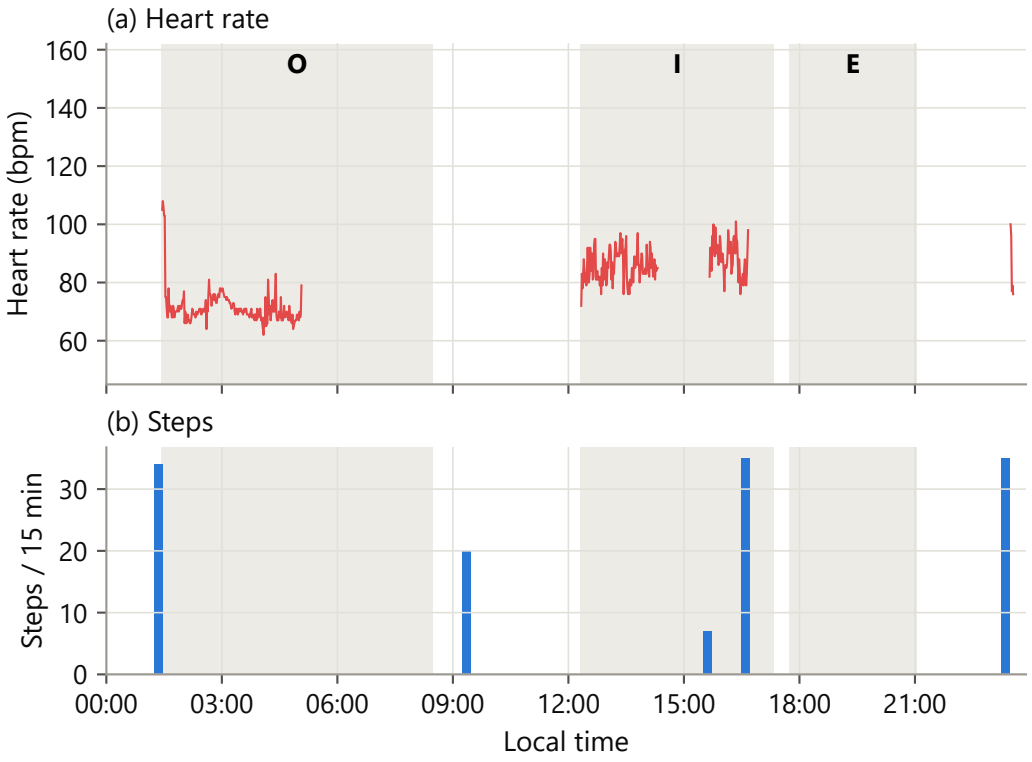


Figure 8: Wrist band over day 4: (a) heart rate (1 sample/min) and (b) steps per 15 min; sleep 01:41–05:08. Grey bands mark the sensor sessions: O bedroom night (01:26–08:28), I indoors, midday and afternoon (12:19–17:20), E indoors, evening (17:43–21:03). Breaks in the heart-rate line are periods with no band data.

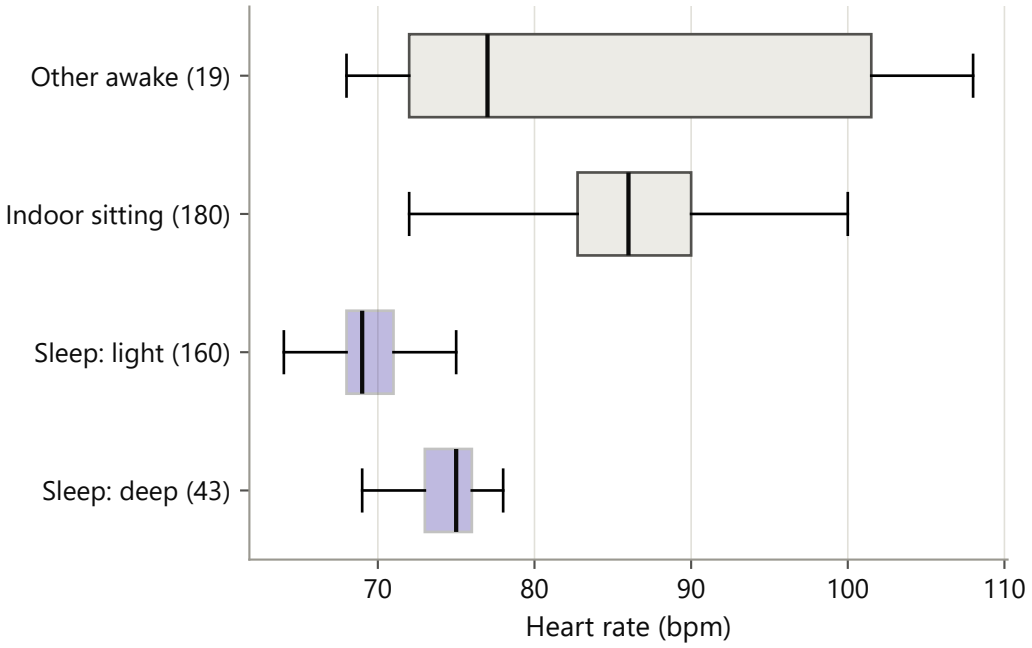


Figure 9: Heart rate by context over the day (minutes per context in brackets). Sleep stages come from the wrist band; awake contexts come from the sensor sessions (gait tests, indoor sitting with the environment node, walking sessions) and all remaining awake minutes. Boxes show the inter-quartile range and median; whiskers extend to 1.5 IQR.

7 Drinking

The bottle log has 1,550 mL in four drinks (Fig. 10): 25 mL at 09:20, 1,000 mL at 11:16, 500 mL at 20:07 and 25 mL at 22:43. Two-thirds of the day's water (1,025 mL) was drunk by 11:16. The afternoon indoors, at corrected 33–37 °C (probably a few degrees less in reality), had no drinks at all, an 8.8-hour gap until 20:07. Compared with days 2–3 (Fig. 14), this is her largest total (599, 1,075 and 1,550 mL) and the first day on which she drank in the morning rather than mainly at night. It is close to the roughly 1.6 L from drinks implied by the EFSA adequate intake for women (2.0 L/day including food). If some drinks went unlogged, these are minimum values.

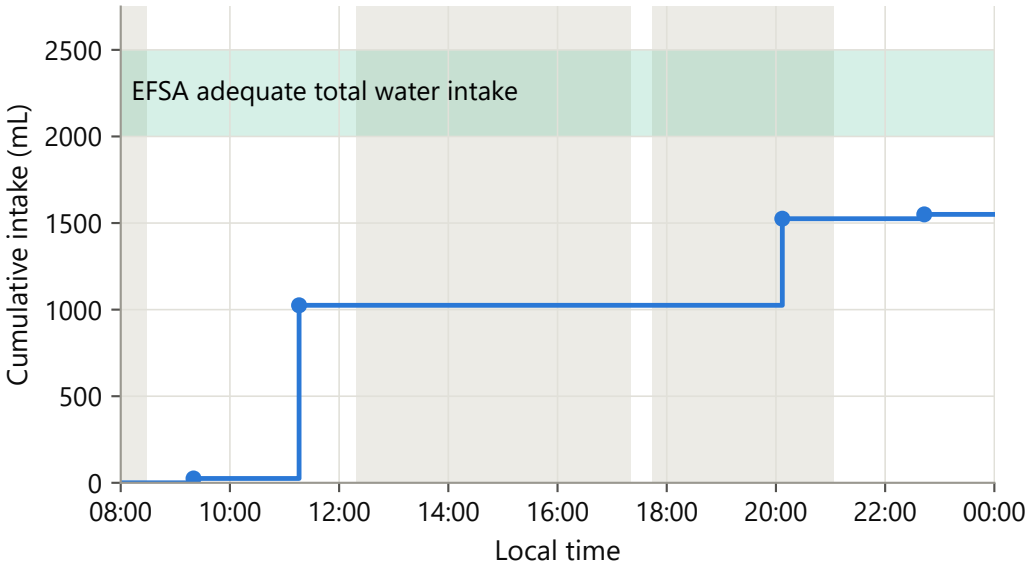


Figure 10: Cumulative fluid intake from the smart-bottle log (4 drinks, 1550 mL). Grey bands mark the sensor sessions; the shaded band is the EFSA adequate intake for total water (2.0–2.5 L/day, including water from food). Longest gap without drinking: 9 h (11:16–20:07).

8 Correlation and covariance

Within the day (Figs. 11 and 12) the strongest correlations are either physical or reflect the day's timeline, not one variable driving another:

- *PM_{10} and $PM_{2.5}$* ($\rho = 0.99$): one source and one sensor, as on every day.
- *Humidity and temperature* ($\rho = -0.97$ overnight, -0.58 indoors): relative humidity falls when the sensor warms the same air; the overnight value is the node's warm-up.
- *Temperature and CO_2* ($\rho = -0.68$ indoors) and *sound and CO_2* ($\rho = 0.53$): the evening had high CO_2, a slightly louder fan floor and lower temperatures, while the midday had low CO_2 and the warm-up. These are two periods differing in several ways, not a link between the variables.
- *Heart rate and the environment*: all $|\rho| \leq 0.32$, based on only 3.5 h of sleep and 3 h awake.

Across days 1–4 (Table 3; covariances and slopes are listed in the file `relationships.csv`), heart rate against steps is still the only heart-rate

relationship that is positive on every day ($\rho = 0.21$, 0.30, 0.28, 0.17; pooled $r = 0.29$, 95 % CI 0.16–0.38; covariance 67 bpm·steps/min; slope 0.18 bpm per step/min). Its verdict changed from "level only" (days 1–3) to "varies" only because day 4 falls below the 0.2 threshold, and day 4 had just 3 minutes with steps. Minute-to-minute changes are again unrelated ($r = -0.04$). Heart rate during sleep is not reliably related to any bedroom variable on the two nights available.

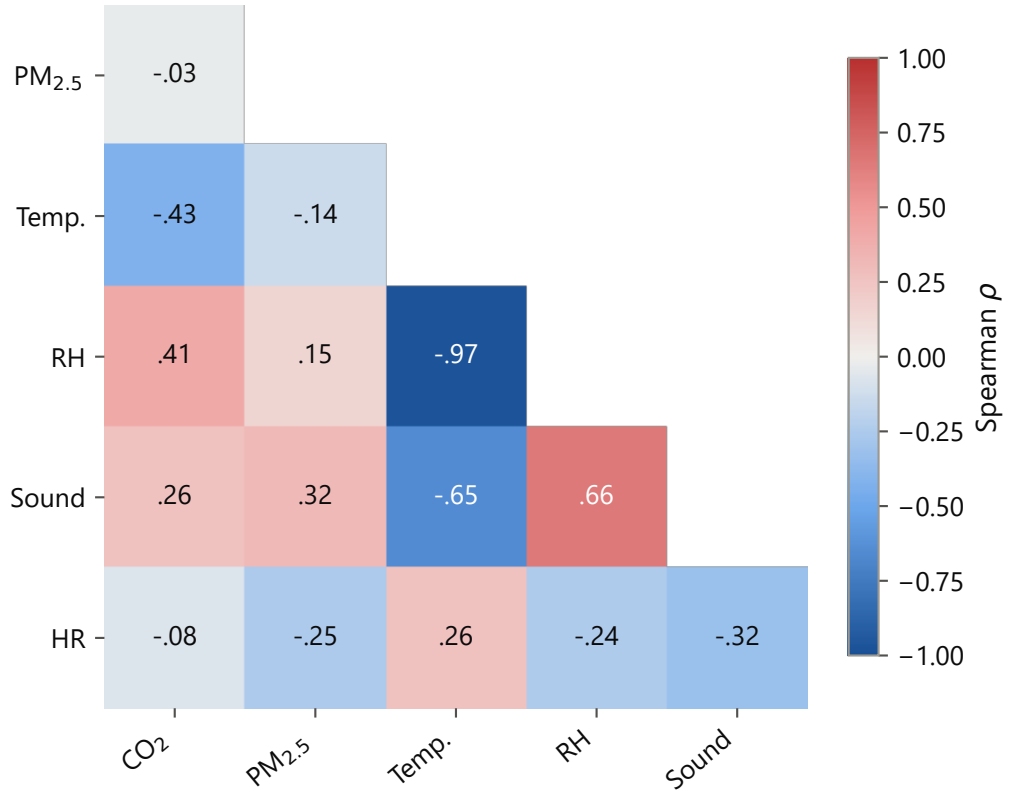


Figure 11: Correlations between bedroom environment and heart rate during sleep (session(s) O, sleep minutes only). Spearman rank correlations of 1-min means ($n = 203$–208 minutes per pair). Minute samples are autocorrelated, so coefficients are descriptive and no p-values are reported.

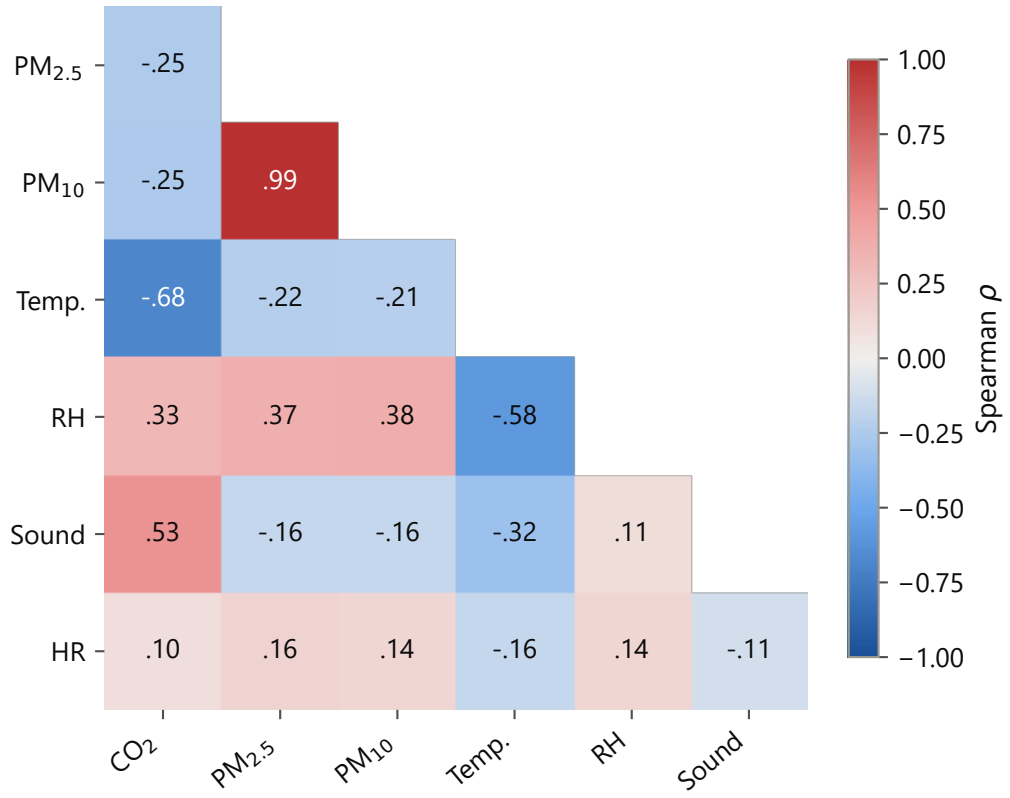


Figure 12: Correlations between indoor environment variables and heart rate during session(s) I, E. Spearman rank correlations of 1-min means ($n = 180$–500 minutes per pair). Minute samples are autocorrelated, so coefficients are descriptive and no p-values are reported.

Table 3: Relationships between per-minute measurements, days 1–4. Pairs are listed as heart rate (or the first variable named) against the second. Per day: Spearman ρ on each day. Level: pooled within-day (repeated-measures) Pearson r with 10-minute block-bootstrap 95 % CI. Change: the same for minute-to-minute changes (*: CI excludes zero). Consistent: same sign with $|\rho| > 0.2$ on every day, both CIs excluding zero and $|r| \geq 0.3$; level only: consistent in levels but not in changes; varies: pooled CI excludes zero but per-day signs or sizes differ. Covariances and regression slopes are given in relationships.csv.

	Per-day ρ						
Pair	D1	D2	D3	D4	Level r (95 % CI)	Change r	Verdict
Heart rate, awake							
Steps	.21	.30	.28	.17	.29 (.16, .38)	−.04	varies
Heart rate, asleep							
Temperature	–	–	−.04	.26	.12 (−.05, .29)	.03	none
CO_2	–	–	.11	−.08	.06 (−.06, .17)	−.03	none
$PM_{2.5}$	–	–	−.25	−.25	−.14 (−.35, .05)	.00	none
Sound	–	–	.04	−.32	−.22 (−.38, −.04)	.00	varies
Environment							
PM_{10} vs $PM_{2.5}$	.92	.99	.98	.98	.98 (.97, .98)	.91*	consistent
RH vs temperature	−.91	.26	−.92	−.73	−.56 (−.74, −.35)	−.61*	varies

9 Days 1–4

Table 4: Spriha's four days (measures that are valid for her). *Incomplete: the band recorded only part of the night. †Band worn for only 6.7 h, all at home.

Measure	Day 1	Day 2	Day 3	Day 4
Walk cadence (steps/min)	116–118	105–108	107–114	–
Stride time (s)	1.02–1.04	1.10–1.14	1.06–1.13	–
Stance, left pressure sensors (%)	62	65	62	–
Foot-tilt range, left / right (°)	51 / 48	55 / 51	55–64 / 46–48	–
Awake heart rate, mean (bpm)	94	96	95	86†
Steps (band)	1,045	1,571	3,731	131†
Heart rate in the rickshaw (bpm)	95	118–122	88–94	no ride
Bedroom $PM_{2.5}$ ($\mu g/m^3$) / sound (dBA)	–	–	69 / 49	57 / 48
Indoor $PM_{2.5}$ ($\mu g/m^3$) / sound (dBA)	77 / 55	37 / 71	–	54 / 60
Upright or straight sitting (%)	–	99	79	–
Fluid logged (mL)	–	599	1,075	1,550
Band sleep record (h)	–	3.2*	2.0*	3.5*
Mean heart rate asleep (bpm)	–	72	70	70

Day 4 adds a home day to three days that each included a short outing and a rickshaw ride. Without insole data, the gait comparison (cadence 105–118 steps/min, stride 1.0–1.1 s, left pressure stance 62–65 %; larger left-foot rotation on all three days) still rests on days 1–3. Indoor $PM_{2.5}$ was between her other indoor days (Fig. 13), and the bedroom was cleaner than on day 3. Her sleeping heart rate has been stable at 70–72 bpm on all three nights recorded. Her logged water intake has risen each day (Fig. 14).

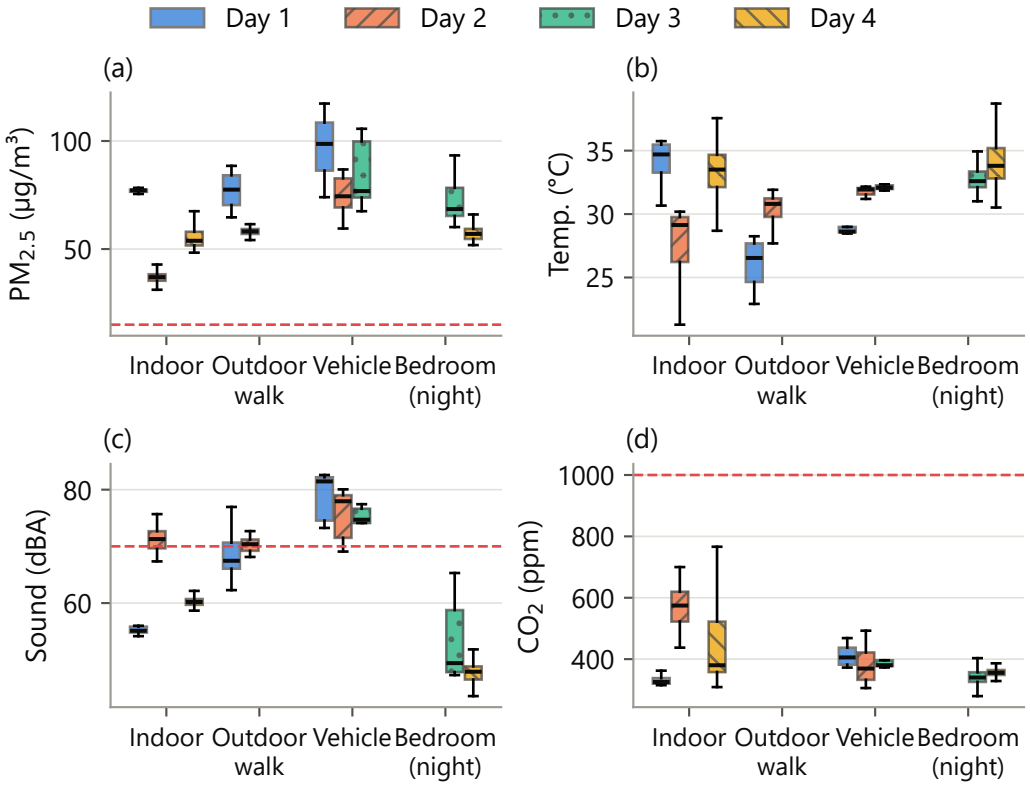


Figure 13: Environment-node readings (1-min means) by setting and day: (a) $PM_{2.5}$, (b) temperature, (c) sound level and (d) CO_2; outdoor CO_2 readings below 400 ppm are excluded as implausible. Dashed lines: WHO $PM_{2.5}$ 24-h guideline, 70 dBA, 1000 ppm. A setting is missing on days without that recording. Temperatures include a -5 °C self-heating correction.

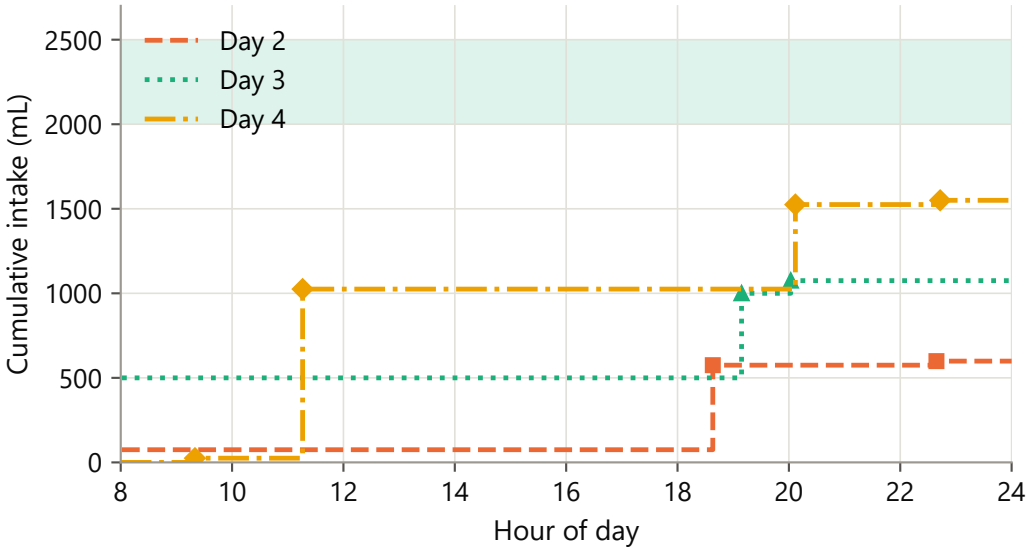


Figure 14: Cumulative fluid intake from the smart-bottle log on each day. The shaded band is the EFSA adequate intake for total water (2.0–2.5 L/day including food).

10 Comparison with participant 1

- **Settings** (Fig. 15, days 1–4): similar indoor $PM_{2.5}$ (median 53 vs 52 $\mu g/m^3$) and indoor sound (61 vs 63 dBA). Her bedroom was quieter (48 vs 62 dBA) and had fewer particles (64 vs 76). Her rooms read much warmer (indoor 33 vs 26 °C, bedroom 33 vs 29 °C), but part of that difference is probably the node's larger self-heating in her recordings (Section 5), so the temperature gap should not be read as fact.
- **Heart rate**: indoors 89 vs 97 bpm, in the bedroom 70 vs 63 bpm, while walking 109 vs 117 bpm.
- **Gait** (Fig. 16, unchanged as day 4 had no insole data): shorter, quicker strides than participant 1 (1.08 vs 1.18 s; 112 vs 102 steps/min), with the same left pressure stance (64 %).
- **Drinking**: her 1.55 L on day 4 is within participant 1's range (0.9–2.2 L per day).

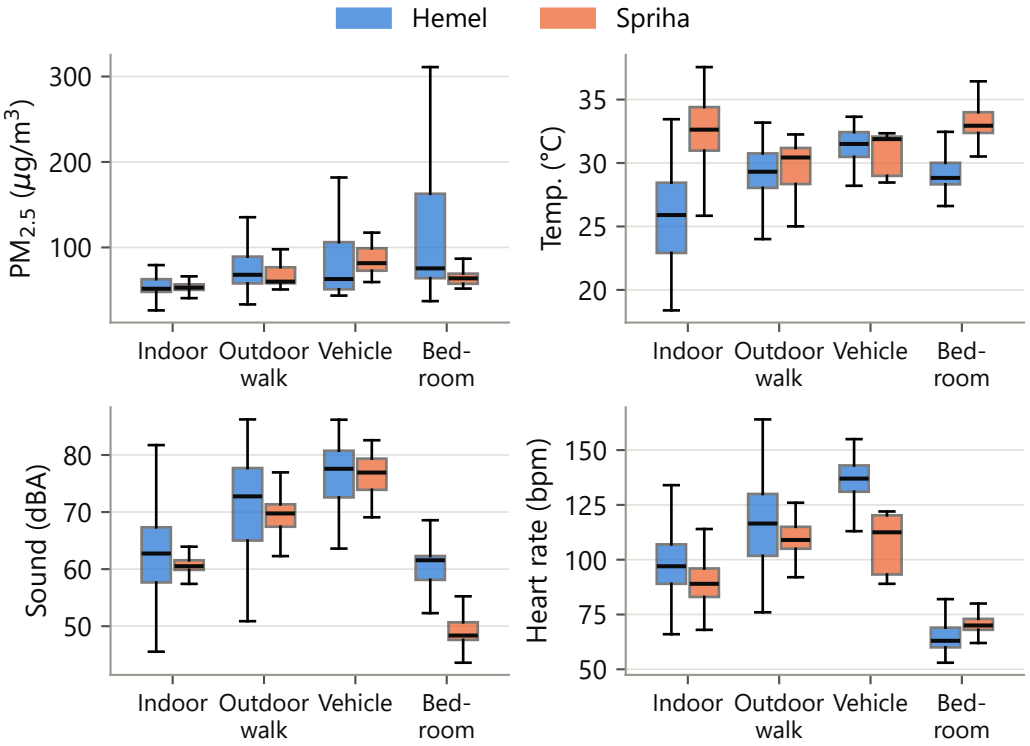


Figure 15: Environment and heart rate by setting for the two participants (1-minute values; Hemel days 1–8; Spriha days 1–4). Temperatures include the −5 °C correction. Spriha's band recorded heart rate every 10 minutes on days 1–2 and every minute from day 3, so her heart-rate boxes rest on fewer minutes.

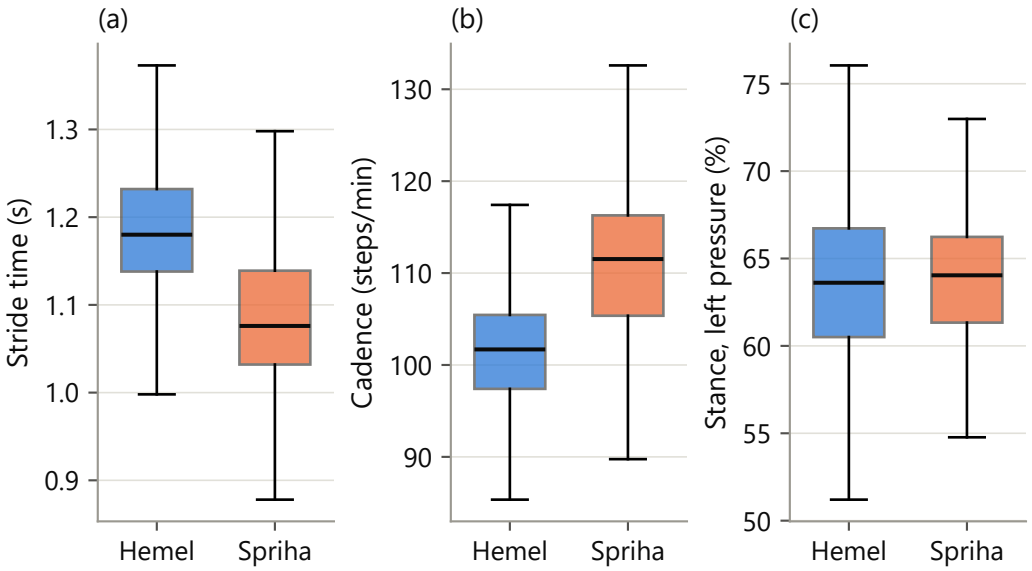


Figure 16: Gait of the two participants in all walks (steady strides; Hemel: days 1–8, 16,161 strides; Spriha: days 1–3, 927 strides). (a) Stride time, (b) cadence and (c) stance phase from the left insole's pressure sensors (the left insole's sensors were original and healthy for both participants).

11 Sensor-system notes

1. **Insoles**: no connection all day in any of the three recordings. It is worth checking the insoles' charge and pairing before the next day; if the phone keeps showing "reconnecting", restarting Bluetooth or the app before recording may help.

2. **Environment-node warm-up**: a 10–12 °C rise after a cold start. Let the node run for 30 minutes before a session, or exclude the first 30 minutes of temperature and humidity. The fixed −5 °C correction is too small for this node and placement; a reference thermometer reading next to the node would allow it to be recalibrated.

3. **Node drop-outs**: the bedroom node stopped at 08:28 (recording to 12:17) and the evening node dropped 10 times after 21:03. Keeping the phone close to the node may help.

4. **Band**: worn for 6.7 h. It was off from about 05:05 to 12:19 and from 16:40 to 23:20, so the day's steps, awake heart rate and sleep are partial. Its sleep-stage labels are doubtful (heart rate higher in "deep" sleep).

5. **GPS indoors** gives fixes scattered over about 60 m; they should not be read as movement.

6. **Notes**: brief notes on when the room was closed and when the air conditioner or fan was switched on would turn the evening-room finding from inference into ground truth, which is useful for the digital twin's context labels.